

PROJECT PRESENTATION

# SMART TRAFFIC MONITORING SYSTEM

An Automated AI & IoT Smart Emission, Noise & Traffic Control System

PROJECT THEME

**Smart Transportation & Environmental Protection**

TEAM NAME

**Green Innovators AP**

SCHOOL NAME

**Sri Gawtham CBSE School, Malkipuram**

TEAM MEMBERS (CLASS 9)

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# Real-World Challenges

Key Traffic & Pollution Issues Affecting Public Safety & Environment

## ☁️ Heavy Vehicle Smoke & Neglected Engines

Vehicle owners skip engine oil maintenance, emitting toxic smoke and carbon particulates on city roads.

## 🔊× Illegal Silencers & Noise Pollution

Aftermarket modified silencers create deafening noise, disturbing residential and hospital zones.

## 🚗 Over-Speeding & Reckless Driving

Excess speed leads to fatal accidents, while existing manual checks fail to monitor roads 24/7.

## 👮× Inefficiency of Manual Traffic Checking

Traffic police cannot manually inspect every smoking or speeding vehicle on busy highways.





# Proposed Solution: Smart Pole

Integrated Multi-Sensor & Optical Computer Vision Framework

01

## AI Optical Smoke Detection

Road-side AI cameras powered by Computer Vision analyze the density and color of exhaust smoke emitting from vehicles in real-time.

02

## Multi-Sensor Traffic Pole

Combines Speed Radar, Decibel Noise Sensors, and Optical Smoke Density Sensors mounted on road gantries to monitor vehicle parameters simultaneously.

03

## Automated Penalty System

When emission or speed limits are breached, ANPR captures the license plate and instantly alerts Traffic Police and the vehicle owner via SMS.

# Hardware & Technology Integration

## Hardware Components & Software Technology Stack

### Hardware Components



#### AI Camera (ANPR)

Captures HD photos of license plates and visual smoke



#### MQ-135 / Optical Gas Sensor

Measures carbon particulate density in exhaust



#### Radar Speed & Acoustic Sensor

Detects vehicle velocity and illegal silencer noise



#### Raspberry Pi 4 / GSM 4G Module

Edge processing microcontroller with cloud connectivity

### Software Stack



#### Python & OpenCV

Real-time video processing and smoke analysis



#### Tesseract OCR / ANPR

Automatic Number Plate Recognition algorithm



#### IoT Cloud Server

Stores violation logs and vehicle records safely

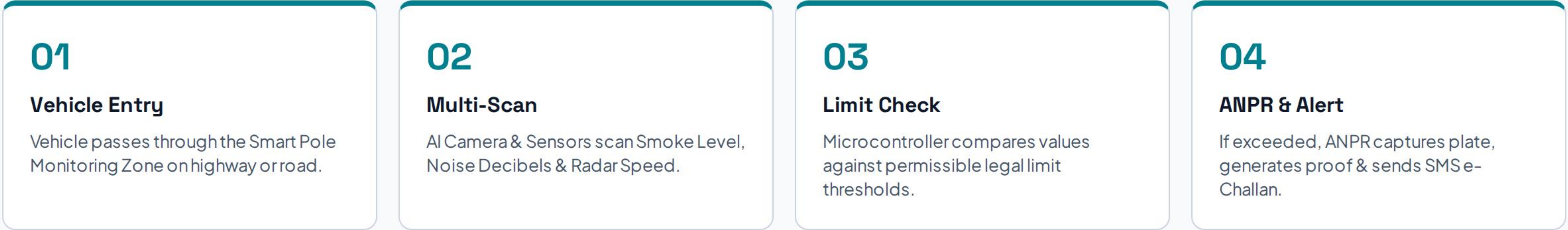


#### Twilio / RTO Gateway API

Triggers instant e-challan SMS to vehicle owners

# Step-by-Step Working Mechanism

Automated Sequence from Vehicle Entry to Penalty Trigger



# Innovative Features

High Accuracy & Scalable Design Features

## Dual Smoke Verification

Uses both gas sensors AND camera optical smoke detection for 99% accuracy against wind dispersion.

## All-in-One Multi-Violation Detection

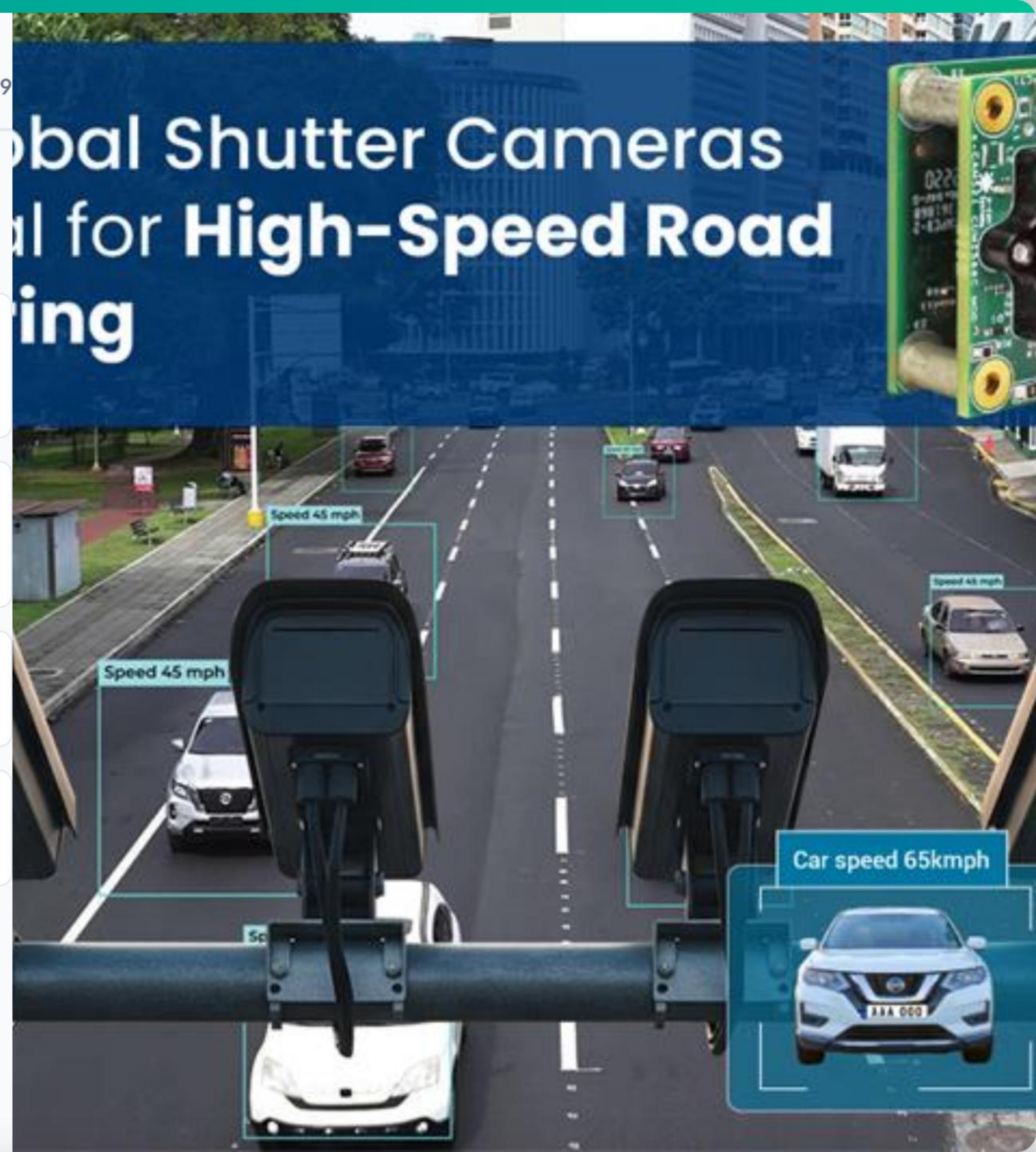
Single camera unit handles Emission, Loud Noise, Over-speeding, and Triple Riding to reduce costs.

## Automated Warning & Tiered Penalty

First violation triggers SMS repair warning; repeated offenses issue heavy RTO fines.

## Solar-Powered Off-Grid Deployment

Equipped with mini solar panels for installation on remote highways without main grid power.



# Budget & Challenge Solutions

Estimated Prototype Cost and Mitigation Strategies

| Component Description          | Cost (INR)     |
|--------------------------------|----------------|
| Raspberry Pi 4 Microcontroller | ₹4,500         |
| HD Wide Angle Camera Module    | ₹1,500         |
| MQ-135 Gas & Sound Sensor Kit  | ₹800           |
| Radar Speed Measurement Sensor | ₹1,200         |
| GSM Module & Battery Unit      | ₹1,500         |
| TOTAL ESTIMATED PROTOTYPE COST | ₹9,500 Approx. |

## Challenge: Heavy Rain / Fog

Disrupting camera view and optical detection in bad weather.

Solution: IR Night Vision + Weatherproof IP66 Casing

## Challenge: High-Speed Vehicles

Fast-moving vehicles slipping past standard camera frame rates.

Solution: High Frame-Rate Shutter Camera (60 FPS)

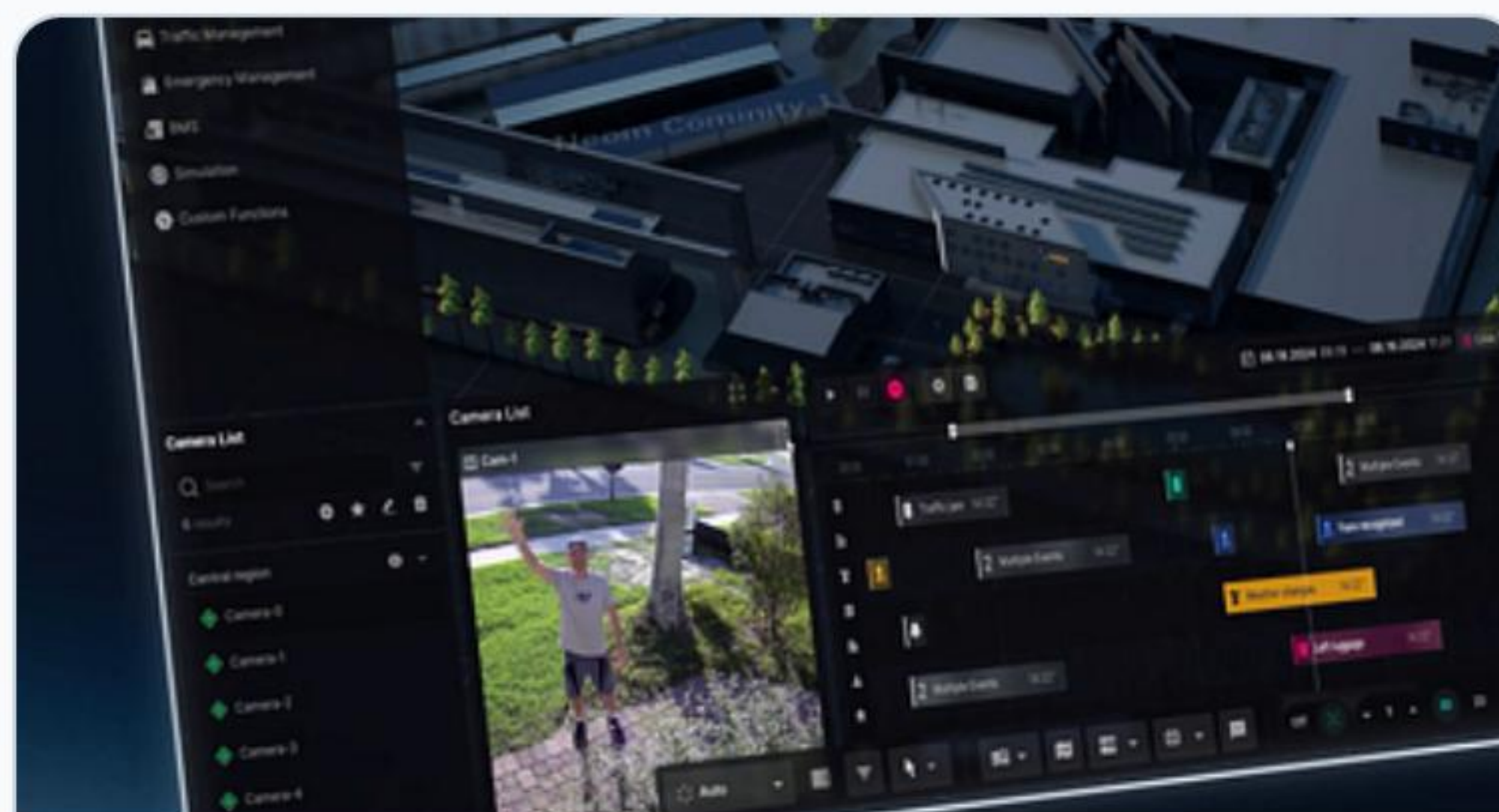
# Environmental & Societal Benefits

Driving Public Health Improvement and Smart Governance



## Pollution Reduction

Curbing vehicular carbon emissions and illegal silencer noise creates healthier urban air and environments.



## Smart City Scalability

Seamlessly integrates into existing Smart City Command Centers, national highway tolls, and signal gantries.



## Zero Corruption & Shift

Automated e-Challan removes manual corruption while encouraging owners to perform timely engine maintenance.

# Project Summary & Future Upgrades

Towards Automated, Affordable & Clean Highway Traffic

## Project Summary

The AI & IoT Automated Traffic Monitoring System offers an affordable, 24/7 automated solution to tackle vehicle smoke emissions, noise pollution, and over-speeding efficiently without manual traffic delays.

**THANK YOU!**

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## Future Upgrades

- ✓ **5G IoT Edge Integration:** Ultra-fast cloud processing and zero latency violation alerts.
- ✓ **Drone-based Aerial Units:** Deploying sensors on mobile drones for highway traffic surveillance.
- ✓ **Central RTO Auto-Sync:** Direct integration with state vehicle databases for instant challan collection.

# Image Sources



[https://www.binbin.tech/wp-content/uploads/2024/10/6458cec11fb3d94d1699be4b\\_shutterstock\\_7480606182.jpg](https://www.binbin.tech/wp-content/uploads/2024/10/6458cec11fb3d94d1699be4b_shutterstock_7480606182.jpg)

Source: [www.binbin.tech](https://www.binbin.tech)

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<https://www.e-consystems.com/blog/camera/wp-content/uploads/2025/06/Why-Global-Shutter-Cameras-Are-Ideal-for-High-Speed-Road-Monitoring.jpg>

Source: [www.e-consystems.com](https://www.e-consystems.com)

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